American International University- Bangladesh (AIUB)
Faculty of Engineering

Final term assignment: 1

Course Name: Introduction to Electrical Circuits **Course Code:** EEE2101

Faculty Name: [redacted] **Semester: Spring 2024 - 25** **Submission Date: 16/06/2025**

Item	COs	POIs	K	P	A	Marks	Obtained Marks
Q1	CO3	P.a.2.C3	K2			6	
Q2	CO3	P.a.2.C3	K2			8	
Q3	CO1	P.a.2.C3	K2			6	
					Total:	20	

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Department:	**CSE**	**Section:**	**Y**

1. Calculate the R.M.S. value of the induced e.m.f. per phase of a 10-pole, 3-phase, 50-Hz alternator with 2 slots per pole per phase and 4 conductors per slot in two layers. The coil span is 150°. The flux per pole has a fundamental component of 0.12 Wb and a 20% third component.

Ans: no: 1

Given, frequency = 50 Hz
3- phase Alternator.
No. of Poles, $P = 10$
Slots per pole per phase, $n = 2$
Total slots $= (2 \times 3 \times 10) = 60$
slots per pole, $m = \frac{60}{10} = 6$
Conductors per slot = 4 (in two layers)
Total conductors $= 4 \times 60 = 240$
Conductors per phase, $Z = \frac{240}{3} = 80$
coil span = 150°
Flux per pole, $\phi = 0.12$ Wb
Third component = 20%.
As, coil span = 150°; Pitch angle $= (180° - 150°) = 30°$
$\therefore \cos\theta = \cos(30°) = 0.866$
For pure sine wave, form factor, $K_f = 1.11$
$\beta = \frac{180°}{6} = 30°$
Each conductor pair makes 1 turn.
Now, $T_c = 240/(2 \times 3) = 40$ turns/ Phase.
Distribution factor will be:

$$K_d = \frac{\sin(n\beta/2)}{n\sin(\beta/2)} = \frac{\sin(30°)}{2\sin(15°)} = 0.966$$

Therefore, RMS. value of Induced E.M.F per Phase,

$$E = 4.44 \times f \times \phi \times T_c \times K_p \times K_d \times K_f$$
$$= 4.44 \times 50 \times 0.12 \times 40 \times 0.966 \times 0.966 \times 1.11$$

$= 1103.75$ V (RMS, per phase).

Therefore, RMS value of the Induced emf per phase $= 1103.75$ V

2. The core of a three phase, 50 Hz, 12000/600 V delta/star, 400 kVA, core-type transformer operates with a flux of 0.045 Wb. Find

a. number of H.V. and L.V. turns per phase.

Ans: no: 2

(a)

We know,

$$E = 4.44 \times f \times \phi \times N$$

$$\text{or, } N = \frac{E}{4.44 \times f \times \phi}$$

For Delta (H.V. side):

$$N = \frac{12000}{4.44 \times 50 \times 0.045}$$

$$= 1201.2 \text{ turns}$$

Given,

$E_{ph} = V_{Line} = 12000\ V$

$f = 50\ Hz$

$\phi = 0.045$

For Star (L.V. side):

$$N = \frac{346.41}{4.44 \times 50 \times 0.045}$$

$$= 34.7 \text{ turns}$$

Given,

For star, $V_{line} = 600V$

$$E_{ph} = \frac{V_{Line}}{\sqrt{3}} = \frac{600}{\sqrt{3}}$$

$$= 346.41\ V$$

Therefore, in HV; number of turns = 1201.2 and for LV; number of turns = 34.7

b. e.m.f. per turn

(b)

We know,

EMF, $E = 4.44 \times f \times \phi$

$= 4.44 \times 50 \times 0.045$

$= 9.99\ V/turn$

Here,

$f = 50$

$\phi = 0.045$

Therefore, EMF per turn $= 9.99\ V$.

c. full load H.V. and L.V. phase-currents.

(c)

We know,

For delta (HV side):

$$I_{Line} = \frac{S}{\sqrt{3} \times V_{Line}}$$

$$= \frac{S}{\sqrt{3} \times 12000}$$

$$= \frac{400,000}{\sqrt{3} \times 12000}$$

$$= 19.24 \text{ A}$$

Apparent power,

$S = 400 \text{ KVA}$

$= 400,000 \text{ VA}$

V_{Line} (For delta)

$= 12000 \text{ V}$

Now,

Phase current, $I_{ph} = \frac{I_{Line}}{\sqrt{3}} = \frac{19.24}{\sqrt{3}}$

$= 11.11 \text{ A}$.

For Star (LV side):

$$I_{Line} = I_{ph} = \frac{400000}{\sqrt{3} \times 600}$$

$$= 384.9 \text{ A}$$

Therefore, full loads H.V. and L.V. Phase currents = 11.11 and 384.9 A.

d. One fourth load H.V. and L.V. phase-currents.

(d)

For delta (HV side):

$$I_{ph(1/4)} = \frac{11.11}{4}$$

$$= 2.78 \text{ A}$$

From 'c' we get,
For HV;
$I_{ph} = 11.11$ A.

For star (LV side):

$$I_{ph(1/4)} = \frac{384.9}{4}$$

$$= 96.23 \text{ A}$$

From 'c' we get;
for LV;
$I_{ph} = 384.9$

Therefore, One-fourth (1/4) load H.V. and L.V. Phase-currents = 2.78 and 96.23 A.

3. A 75-kW, 3-ϕ, Y-connected, 50-Hz, 440-V cylindrical rotor synchronous motor operates at rated condition with 0.8 p.f. leading. The motor efficiency excluding field and stator losses, is 95% and Xs=2.5 Ω. Calculate:

(i) mechanical power developed

Ans: no: 3

(i)

We know,

$P_m = P_{in}$.

We know,

$$P_{in} = \frac{P_{out}}{\eta} = \frac{75 \times 10^3}{0.95} = 78{,}950\,W$$

Therefore mechanical Power developed,

$P_m = 78{,}950\,W$.

(ii) armature current

(ii)

Given,

Line voltage, $V_{Line} = 440\,V$

Now,

Phase voltage;

$$V_{ph} = \frac{V_{line}}{\sqrt{3}} = \frac{440}{\sqrt{3}} = 254.03$$

Here,

Phase = $3-\phi$

$$\therefore P_{in} = \sqrt{3} \cdot V_{Line} \cdot I_{Line} \cdot \cos\phi$$

$$\text{or, } I_{Line} = \frac{P_{in}}{\sqrt{3} \times V_{Line} \times \cos\phi}$$

From 'a' we get, input power, $P_{in} = 78,950\ W$

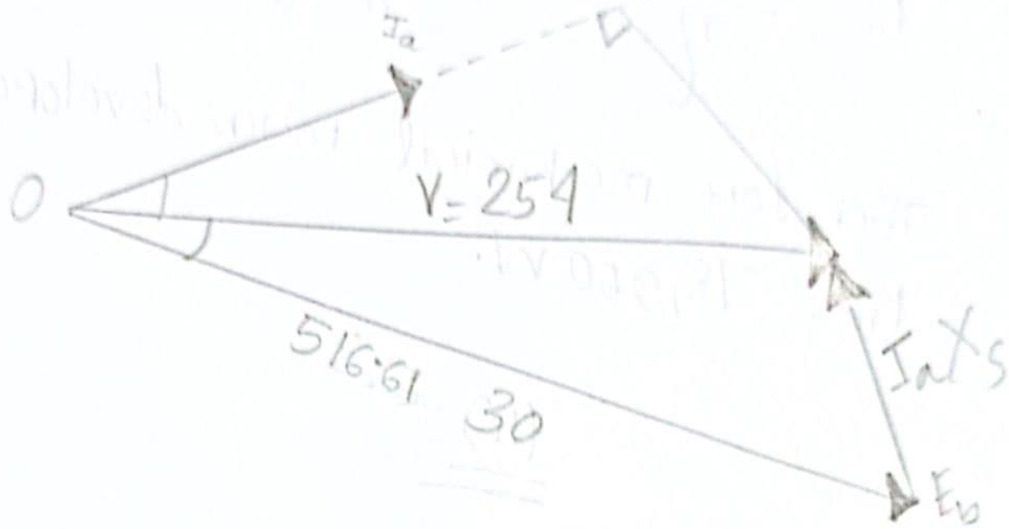
$$\text{or, } I_{Line} = \frac{78,950}{\sqrt{3} \times 440 \times 0.8}$$

$$\text{or, } I_{Line} = 129.41\ A$$

Therefore, Armature current, $I_{Line} = 129.41\ A$.

(iii) back e.m.f.

(iii)



Here,

Applied voltage/phase $= \frac{440}{\sqrt{3}} = 254\,V.$

Let,

$V = 254\angle 0^\circ$ (As shown in the above figure).

Now,

$V = E_b + jI_aX_s$

or, $E_b = V - jI_aX_s$

or, $E_b = 254\angle 0^\circ - 129\angle 36.9^\circ \times 2.5\angle 90^\circ$

or, $E_b = 516.61\angle -30^\circ$

Therefore, back emf; $E_b = 516.61\,V\angle -30^\circ$

(iv) power angle and

(IV)

From (III);

we get;

Back EMF, $E_b = 516.61\ V\angle -30^\circ$

$\therefore \alpha = -30^\circ$

Therefore power angle, $\alpha = -30^\circ$.

(v) maximum or pull-out torque of the motor

(v)

As we know that,

Pull-out torque occurs if $\alpha = 90^\circ$

We also know,

Maximum Power, $P_m = 3\frac{E_b V}{X_s}\sin\delta$

$= 3\ \frac{256 \times}{}$

$= 3 \times \frac{516.61 \times 254}{2.5} \times \sin 90^\circ$

$= 157,462.73$

Now, for pull-out torque:

$$\tau_{max} = \frac{P_{max}}{\omega} = \frac{157,462.73}{1500 \times \left(\frac{2\pi}{60}\right)}$$

$= 1002.44\ N\text{-}m$

Therefore, maximum or pull-out torque = 1002.44 N-m.